

BARRETT RUTH

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EDUCATION

University of Virginia

Bachelor's of Computer Science

Charlottesville, VA

Aug. 2022–May 2026

- Coursework: Advanced Algorithms, Operating Systems, Algorithmic Economics, Group Theory GPA: 3.96/4.00
- Activities: ICPC, HooHacks 2x Winner

EXPERIENCE

IMC

Software Engineer

August 2026

Chicago, IL

- Incoming C++ execution engineer

Neovim

Core Maintainer

May 2026–Present

Remote

- One of 27 maintainers of the world's most admired editor (3M+ users)

Ramp

Backend Software Engineering Intern

September–December 2025

New York City, NY

- Built bulk fraud dispute processor, saving $\approx$ \$100,000/yr and $\approx$ 10,000 CX hours/yr
- Stabilized $\approx$ 10% of backend tests by removing SQLAlchemy session-sharing in Temporal tests

DRW

Software Developer Intern

June–August 2025

Houston, TX

- Built “Exchange Equivalence,” a portfolio risk hedging tool for FTR trade exposures
- Developed “Constraint Exposure” FTR arbitrage tooling suite
- Optimized dynamic spanning tree across all market simulations, saving $\approx$ \$10,000/mo

UVA Insight Computer Architecture Lab

Hardware Profiling Research Assistant

January–May 2025

Charlottesville, VA

- Profiled heterogeneous memory systems of NVIDIA GH200 Superchip for large-scale GEMM

Visa

Software Engineering Intern

May–August 2024

Austin, TX

- Built “GenAI Support Assist,” automating global authentication issue resolution, reducing team debugging by 8 hours/week; selected by SVP of Payment Solutions
- Only intern to deploy to CloudView (cloud platform); integrated GenAI APIs (Ada-002, GPT) for real-time production polling, VectorDB/MySQL storage, and authenticated support frontend

PROJECTS

Neovim Tooling | *Lua, C, libuv, Plenary*

May 2025–Present

- Core maintainer of none-ls, Neovim's largest (2.9k+ stars, 10k users) LSP integration framework
- Built guard, Neovim's first async formatting and linting plugin (6k users) via Lua coroutines

Cavalier Autonomous Racing | *ROS, C++, CMake*

April 2024–May 2026

- Refactored ROS2 state machine for emergency racing scenarios with in-house YAML validator
- Migrated GUI to Qt C++, increasing telemetry throughput 25% via async monitoring

theCourseForum | *Django, PostgreSQL, jQuery, AWS*

August 2022–May 2025

- Led largest university course analytics platform serving 20,000+ students (85% of student body)
- Replaced Elasticsearch with trigram/reverse-index search, saving \$150/mo

TECHNICAL SKILLS

Languages: C, C++, Python, Lua, Nix, SQL, TypeScript, JavaScript, Java, Shell, HTML/CSS

Technologies/Frameworks: Linux, Git, Docker, PostgreSQL, Redis, Protobuf, React, Next.js, Node.js, AWS, GCP, Jenkins